Overall Intro and Sequent Calculus 1

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LMFI Proof Theory Course
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Course Context: 4 Landmark Theorems in Proof Theory

- Gentzen's cut-elimination theorem for sequent calculus
  - Establishes consistency of several deduction systems
  - Implies disjunction and existence properties of intuitionistic systems
- Gödel's completeness theorem for first-order logic
  - Establishes the connection of syntactic provability and semantic validity
  - Implies semantic cut-elimination if used for Kripke semantics
- Martin-Löf's proof of the Church Rosser theorem for pure lambda calculus
  - Establishes the well-behavedness of lambda terms as programs
  - Implies the uniqueness of normal forms for beta reduction
- Tait's proof of the strong normalisation theorem for simply typed lambda calculus
  - Establishes that every reduction sequence ends in a normal form
  - Implies the consistency of systems seen through the Curry-Howard correspondence

(Tentative) Course Overview

- Sequent Calculus
  - Lectures on Sep 22 and Sep 29 (Alexis)
  - TD on Sep 25
- Natural Deduction
  - Lectures on Oct 6, Oct 13 and Oct 20
  - TDs on Oct 9 and Oct 23
- Untyped Lambda Calculus
  - Lectures on Oct 27, Nov 10 and Nov 17
  - TDs on Nov 6 and Nov 20
- Simply Typed Lambda Calculus
  - Lectures on Nov 24, Dec 1 and Dec 8
  - TD on Dec 4

Course Exams

There will be two exams:

- Midterm (Nov 3): covering the first half of the course
- Endterm (week of Dec 14): covering the whole course

The final grade will be better of the following two options:

- The average grade of both the midterm and endterm exams
- The grade of the endterm exam alone

Course Tips

- Ask questions during the lectures
- Practice exercises in Lisi's TDs every other Friday
- Discuss with other students on Discourse
- Study at home, especially with the recommended textbooks:
  - J.-Y. GIRARD, Y. LAFONT & P. TAYLOR : Proofs and Types (Cambridge University Press, 1989, disponible sur la page de P. Taylor).
  - A.S. TROELSTRA & H. SCHWICHTENBERG: Basic Proof Theory (Cambridge University Press, 2000).

Perspective: Natural Deduction (ND)

In later lectures, we study Gentzen's natural deduction system:[1]

$$\ldots \qquad \frac{A \in \Gamma}{\Gamma \vdash A}\ \mathsf{AX} \qquad \frac{\Gamma, A \vdash B}{\Gamma \vdash A \to B}\ \mathsf{II} \qquad \frac{\Gamma \vdash A \to B \quad \Gamma \vdash A}{\Gamma \vdash B}\ \mathsf{IE} \qquad \ldots$$

Very natural formalisation of logical reasoning:

- Justification of logical connectives is modelled by introduction rules
- Usage of logical connectives is modelled by elimination rules

ND comes in intuitionistic and classical flavours, depending on how falsity is eliminated:

$$\ldots \qquad \frac{\Gamma \vdash_i \bot}{\Gamma \vdash_i A}\ \mathsf{EXP} \qquad \frac{\Gamma, \neg A \vdash_c \bot}{\Gamma \vdash_c A}\ \mathsf{DNE} \qquad \ldots$$

Recall that we treat $\neg A$ as abbreviation for $A \to \bot$ and that $\vdash_c A \vee \neg A$ follows.

[1] Gentzen, G. Untersuchungen über das logische Schließen. I. Math Z 39, 176–210 (1935).

Motivation: Decidability and Related Properties

To check if $\Gamma \vdash A$, can we just systematically try out the rules backwards?

$$\dfrac{\dfrac{\dfrac{\dfrac{}{B \vdash B} \quad \dfrac{\vdots}{B \vdash C}}{B \vdash B \wedge C}\,\mathrm{CI}}{B \vdash A \vee (B \wedge C)}\,\mathrm{DI}_2}{\vdash B \to A \vee (B \wedge C)}\,\mathrm{II}_2$$

Leads nowhere as (IE) is always applicable and introduces completely new formulas!

Other properties, also blocked by the (IE) rule...

- Consistency: is $\vdash \bot$ derivable?
- Disjunction property: if $\vdash_i A \vee B$, can we conclude either $\vdash_i A$ or $\vdash_i B$?

We want an analytic deduction system with a subformula property!

The Classical Sequent Calculus (LK)[2]

Structure derivations $\Gamma \Rightarrow \Delta$ by symmetrically decomposing formulas on either side:

$$\frac{\Gamma \cap \Delta \neq \emptyset}{\Gamma \Rightarrow \Delta}\ \mathsf{AX} \qquad \frac{}{\Gamma, \bot \Rightarrow \Delta}\ \mathsf{BL} \qquad \frac{}{\Gamma \Rightarrow \top, \Delta}\ \mathsf{TR}$$

$$\frac{\Gamma \Rightarrow A, \Delta \quad \Gamma, B \Rightarrow \Delta}{\Gamma, A \to B \Rightarrow \Delta}\ \mathsf{IL} \qquad \frac{\Gamma, A \Rightarrow B, \Delta}{\Gamma \Rightarrow A \to B, \Delta}\ \mathsf{IR}$$

$$\frac{\Gamma, A, B \Rightarrow \Delta}{\Gamma, A \wedge B \Rightarrow \Delta}\ \mathsf{CL} \qquad \frac{\Gamma \Rightarrow A, \Delta \quad \Gamma \Rightarrow B, \Delta}{\Gamma \Rightarrow A \wedge B, \Delta}\ \mathsf{CR}$$

$$\frac{\Gamma, A \Rightarrow \Delta \quad \Gamma, B \Rightarrow \Delta}{\Gamma, A \vee B \Rightarrow \Delta}\ \mathsf{DL} \qquad \frac{\Gamma \Rightarrow A, B, \Delta}{\Gamma \Rightarrow A \vee B, \Delta}\ \mathsf{DR}$$

Directly obtain consistency, as there is no rule applicable for $\Rightarrow \bot$!

[2] Alternative formulations in Girard's and Troelstra/Schwichtenberg's textbooks.

Some Alternative Formulations

If Γ and Δ are treated as multi-sets, we need explicit structural rules:

$$\frac{\Gamma, A, A \Rightarrow \Delta}{\Gamma, A \Rightarrow \Delta}\ \mathrm{cL} \qquad \frac{\Gamma \Rightarrow A, A, \Delta}{\Gamma \Rightarrow A, \Delta}\ \mathrm{cR} \qquad \frac{\Gamma \Rightarrow \Delta}{\Gamma, A \Rightarrow \Delta}\ \mathrm{wL} \qquad \frac{\Gamma \Rightarrow \Delta}{\Gamma \Rightarrow A, \Delta}\ \mathrm{wR}$$

In such a setting, the axiom rule is often replaced by $A \Rightarrow A$ or even $p \Rightarrow p$.

Rules with two premises can also be formulated in a multiplicative way:

$$\frac{\Gamma \Rightarrow A, \Delta \quad \Gamma', B \Rightarrow \Delta'}{\Gamma, \Gamma', A \to B \Rightarrow \Delta, \Delta'}\ \mathrm{IL} \qquad \frac{\Gamma \Rightarrow A, \Delta \quad \Gamma' \Rightarrow B, \Delta'}{\Gamma, \Gamma' \Rightarrow A \wedge B, \Delta, \Delta'}\ \mathrm{CR} \qquad \frac{\Gamma, A \Rightarrow \Delta \quad \Gamma', B \Rightarrow \Delta'}{\Gamma, \Gamma', A \vee B \Rightarrow \Delta, \Delta'}\ \mathrm{DL}$$

These rules are overall equivalent but interact differently with structural rules and cuts.

LK is indeed classical!

Fact

In LK, $\Rightarrow A \vee \neg A$ is derivable for arbitrary A.

Proof.

$$\dfrac{\dfrac{\dfrac{}{A \Rightarrow A, \bot}\,\mathsf{AX}}{\Rightarrow A, \neg A}\,\mathsf{IR}}{\Rightarrow A \vee \neg A}\,\mathsf{DR}$$

□

The Intuitionistic Sequent Calculus (LJ)[3]

To obtain the intuitionistic system, restrict Δ to contain at most one formula:

$$\frac{\Gamma \cap \Delta \neq \emptyset}{\Gamma \Rightarrow \Delta}\ \mathsf{AX} \qquad \frac{}{\Gamma, \bot \Rightarrow \Delta}\ \mathsf{BL} \qquad \frac{}{\Gamma \Rightarrow \top}\ \mathsf{TR}$$

$$\frac{\Gamma \Rightarrow A \quad \Gamma, B \Rightarrow \Delta}{\Gamma, A \to B \Rightarrow \Delta}\ \mathsf{IL} \qquad \frac{\Gamma, A \Rightarrow B}{\Gamma \Rightarrow A \to B}\ \mathsf{IR}$$

$$\frac{\Gamma, A, B \Rightarrow \Delta}{\Gamma, A \wedge B \Rightarrow \Delta}\ \mathsf{CL} \qquad \frac{\Gamma \Rightarrow A \quad \Gamma \Rightarrow B}{\Gamma \Rightarrow A \wedge B}\ \mathsf{CR}$$

$$\frac{\Gamma, A \Rightarrow \Delta \quad \Gamma, B \Rightarrow \Delta}{\Gamma, A \vee B \Rightarrow \Delta}\ \mathsf{DL} \qquad \frac{\Gamma \Rightarrow A}{\Gamma \Rightarrow A \vee B}\ \mathsf{DR}_1 \qquad \frac{\Gamma \Rightarrow B}{\Gamma \Rightarrow A \vee B}\ \mathsf{DR}_2$$

Observe disjunction property, as for $\Rightarrow A \vee B$ only (DR_1) and (DR_2) are applicable!

[3] Alternative formulations in Girard's and Troelstra/Schwichtenberg's textbooks.

LJ is indeed intuitionistic!

Fact

In LJ, $\Rightarrow A \vee \neg A$ is not derivable for arbitrary A.

Proof.

If we for instance choose an atomic formula p, any attempt to derive $p \vee \neg p$ fails:

$$\dfrac{\dfrac{\lightning}{\Rightarrow p}}{\Rightarrow p \vee \neg p} \qquad \dfrac{\dfrac{\dfrac{\lightning}{p \Rightarrow \bot}}{\Rightarrow \neg p}}{\Rightarrow p \vee \neg p}$$

□

So while in ND the rule for the elimination of falsity controls the flavour, for sequent calculi it is a consequence of structural constraints!

Decidability via Naive Proof Search

Its analytic structure yields a systematic way to decide derivability in sequent calculus:

Theorem (Decidability)

There is an algorithm checking whether or not $\Gamma \Rightarrow \Delta$.

Proof Sketch.

1. Freely apply rules backwards, for LJ backtracking might be necessary.
2. If no more rule is applicable, check if (AX) can solve the goal.
3. Termination: with every rule application at least one formula is decomposed.
4. Correctness: either yields a derivation or a refuting assignment. □

Now that we have established all desired properties for sequent calculus, we simply transport them over to ND by showing the systems equivalent!

Equivalence to ND

It will be a nice exercise to show that $\Gamma \Rightarrow \Delta$ implies $\Gamma \vdash \bigvee \Delta$...
Conversely, one gets stuck immediately as reasoning in sequent calculus is too local:

$$\frac{\Gamma \Rightarrow A \to B \quad \Gamma \Rightarrow A}{\Gamma \Rightarrow B} \; ?$$

The solution is to show that there is a hidden but very powerful reasoning principle:

Theorem (Cut Admissibility)

If $\Gamma \Rightarrow C, \Delta$ *and* $\Gamma, C \Rightarrow \Delta$, *then* $\Gamma \Rightarrow \Delta$.

Proof.

This will be the technical content of the next lecture to follow... □

Corollary

If $\Gamma \vdash A$ *then* $\Gamma \Rightarrow A$, *so together with the other direction ND inherits all the desired properties from sequent calculus.*

Rules for Quantifiers

For quantified formulas $\forall x.\, A$ and $\exists x.\, A$, classical rules can be given as follows:

$$\frac{\Gamma, A[t/x] \Rightarrow \Delta}{\Gamma, \forall x.\, A \Rightarrow \Delta}\ \text{AL} \qquad \frac{\Gamma \Rightarrow A[y/x], \Delta}{\Gamma \Rightarrow \forall x.\, A, \Delta}\ \text{AR} \qquad \frac{\Gamma, A[y/x] \Rightarrow \Delta}{\Gamma, \exists x.\, A \Rightarrow \Delta}\ \text{EL} \qquad \frac{\Gamma \Rightarrow A[t/x], \Delta}{\Gamma \Rightarrow \exists x.\, A, \Delta}\ \text{ER}$$

In the (AR) and (EL) rules, y must not occur free in Γ and Δ.

In the intuitionistic version, the right rules are again severely restricted:

$$\frac{\Gamma, A[t/x] \Rightarrow \Delta}{\Gamma, \forall x.\, A \Rightarrow \Delta}\ \text{AL} \qquad \frac{\Gamma \Rightarrow A[y/x]}{\Gamma \Rightarrow \forall x.\, A}\ \text{AR} \qquad \frac{\Gamma, A[y/x] \Rightarrow \Delta}{\Gamma, \exists x.\, A \Rightarrow \Delta}\ \text{EL} \qquad \frac{\Gamma \Rightarrow A[t/x]}{\Gamma \Rightarrow \exists x.\, A}\ \text{ER}$$

Observe existence property, as for $\Rightarrow \exists x.\, A$ only (ER) for concrete $A[t/x]$ is applicable!